

Special Triangle Congruence Theorems and Related Theorems

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Right Triangle Congruence Theorems
provide shortcuts for proving congruence of right
triangles: **HL**, **LL**, **HA**, and **LA**.

HL — Hypotenuse-Leg

If the hypotenuse and one leg of a right triangle are congruent to those of another right triangle, then the triangles are congruent.

LL — Leg-Leg

If both legs of a right triangle are congruent to both legs of another right triangle, then the triangles are congruent.

HA — Hypotenuse-Acute Angle

If the hypotenuse and an acute angle of a right triangle are congruent to those of another right triangle, then the triangles are congruent.

LA — Leg-Acute Angle

If a leg and an acute angle of a right triangle are congruent to the corresponding leg and acute angle of another right triangle, then the triangles are congruent.

Right Triangle Shortcuts at a Glance

HL — hypotenuse, leg

LL — leg, leg

HA — hypotenuse, acute angle

LA — leg, acute angle

All four apply only to **right triangles**.

Isosceles Triangle Theorem

If two sides of a triangle are congruent, then the angles opposite those sides are congruent.

In $\triangle PQR$, if $PQ = PR$, then $\angle Q = \angle R$.

Converse of the Isosceles Triangle Theorem

If two angles of a triangle are congruent, then the sides opposite those angles are congruent.

In $\triangle PQR$, if $\angle Q = \angle R$, then $PQ = PR$.

Perpendicular Bisector Theorem

Any point on the perpendicular bisector of a segment is equidistant from the endpoints of the segment.

If P lies on the perpendicular bisector of AB ,
then $PA = PB$.

Converse of the Perpendicular Bisector Theorem

If a point is equidistant from the endpoints of a segment, then it lies on the perpendicular bisector of the segment.

If $PA = PB$, then P lies on the perpendicular bisector of AB .

Midline (Midsegment) Theorem

The segment joining the midpoints of two sides of a triangle is parallel to the third side and has one-half its length.

If D and E are midpoints of AB and AC in $\triangle ABC$, then

$$DE = \frac{1}{2} BC$$

Examples

Example 1

In isosceles $\triangle ABC$, $AB = AC$. If $\angle B = 50^\circ$, find $\angle C$.

Answer: By the Isosceles Triangle Theorem, the angles opposite the congruent sides are congruent:

$$\angle C = \angle B = 50^\circ$$

Example 2

Point P lies on the perpendicular bisector of segment AB . If $PA = 20$ cm, find PB .

Answer: By the Perpendicular Bisector Theorem,

$$PB = PA = 20 \text{ cm}$$

Example 3

In $\triangle ABC$, D and E are midpoints of AB and AC . If $BC = 18$ cm, find DE .

Answer: By the Midline Theorem,

$$DE = \frac{1}{2}(18) = 9 \text{ cm}$$

Practice Exercises

Practice 1

Two right triangles have hypotenuses measuring 15 cm and corresponding legs measuring 9 cm. Which right triangle congruence theorem proves the triangles congruent?

Practice 2

Right triangles $\triangle ABC$ and $\triangle DEF$ have congruent hypotenuses and one pair of corresponding acute angles congruent. Identify the right triangle congruence theorem that applies.

Practice 3

In isosceles $\triangle PQR$, $PQ = PR$. If $\angle Q = 52^\circ$, find $\angle R$ and $\angle P$.

Practice 4

In $\triangle XYZ$, $\angle X = \angle Z$. If $XY = 13$ cm, determine the length of YZ .

Practice 5

Segment AB has midpoint M . Point P lies on the perpendicular bisector of AB . If $PA = 18$ cm, find PB .

Practice 6

Point Q is equidistant from the endpoints of segment CD . What can you conclude about the location of Q ?

Practice 7

In $\triangle ABC$, D and E are the midpoints of AB and AC , respectively. If $BC = 24$ cm, find DE .

Practice 8

In $\triangle LMN$, XY joins the midpoints of LM and LN . If $XY = 11$ cm, determine MN .

Practice 9

A triangle has sides 14, 14, and 20. Determine the measures of the two base angles if the vertex angle measures 68° .

Activities

Activity 1

Two right triangles have hypotenuses measuring 17 cm and corresponding legs measuring 8 cm. Which right triangle congruence theorem proves the triangles congruent?

Activity 2

Right triangles $\triangle GHI$ and $\triangle JKL$ have congruent hypotenuses and one pair of corresponding acute angles congruent. Identify the right triangle congruence theorem that applies.

Activity 3

In isosceles $\triangle ABC$, $AB = AC$. If $\angle B = 61^\circ$, find $\angle C$ and $\angle A$.

Activity 4

In $\triangle DEF$, $\angle D = \angle F$. If $DE = 18$ cm, determine the length of EF .

Activity 5

Segment XY has midpoint N . Point R lies on the perpendicular bisector of XY . If $RX = 21$ cm, find RY .

Activity 6

Point T is equidistant from the endpoints of segment MN . What can you conclude about the location of T ?

Activity 7

In $\triangle PQR$, S and T are the midpoints of PQ and PR , respectively. If $QR = 30$ cm, find ST .

Activity 8

In $\triangle ABC$, DE joins the midpoints of AB and AC . If $DE = 16$ cm, determine BC .

Activity 9

A triangle has sides 16, 16, and 24. Determine the measures of the two base angles if the vertex angle measures 74° .

Thank you!